A proof of the conjectured order of the recurrence for OEIS A274957

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1 The conjecture

The Online Encyclopedia of Integer Sequences records A274957 as

> *Number of nX6 0..2 arrays with no element equal to any value at offset (-1,-1) (-2,0) or (0,-2) and new values introduced in order 0..2*

and carries in its Formula section the line:

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Empirical recurrence of order 81 (see link above)
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That line carries no separate signature; the entry itself is by R. H. Hardin (Jul 12 2016).

The word *Empirical* — equivalently *Conjecture* — is the entry's own: the statement is recorded as fitted to the published terms, and no proof is given there. As of revision 4, last modified Jul 12 2016, the entry records no proof and links to none, so the statement is open. This note settles it.

The entry states only that the sequence satisfies a linear recurrence of order 81 and refers to a linked file for its coefficients, which this note does not use. What is proved below is the corresponding exact statement: the least order of a linear recurrence the sequence eventually satisfies, and the exact index beyond which it holds.

2 The object

The name is read as follows. The reading is not a guess: it is pinned against the entry's own published terms, and against an independent enumeration, before anything is proved (Section 7).

Let A be an $n' \times 6$ array over $\{0, \dots, 2\}$ with $n' = n$ rows.

The entry's offsets are (-1,-1), (-2,0), (0,-2). The condition is

$$A[i][j] \neq A[i+\delta][j+\varepsilon] \quad \text{for every listed } (\delta, \varepsilon) \text{ landing inside the array.}$$

An offset and its negative say the same thing, so the offsets are normalised here to point upwards or, in the same row, leftwards: (-2,0), (-1,-1), (0,-2).

The entry's further clause, that the new values $0, \ldots, 2$ are introduced in order, means that reading the array in row-major order the first occurrences of the values happen in increasing order: a value v may appear only once every smaller value has already appeared.

$a(n)$ is the number of such arrays. The entry's offset is 1, so its term of index n is the count for n rows.

3 The count is a walk count

Every offset reaches at most 2 rows upwards, so the condition on a row is decided by that row together with the 2 rows above it. Take as state the last 2 rows, together with a counter recording how many of the values $0, \ldots, 2$ have already been introduced — that counter is all the labelling clause needs, since a value may be used exactly when it is smaller than the counter, or equal to it, in which case the counter advances.

Appending a row is a transition, an array of n' rows is a walk of length n' from the empty state, and every array arises from exactly one walk. So $a(n) = w^{\mathsf{T}} M^{n'} f$ and the count is C-finite.

4 The degree bound, derived

The reachable part of the digraph has 3917 states, 3917 after trimming. Identifying states with the same number of completions of every length, and then the mirror identification on the edges coming in, leaves $S = 110$ blocks, and the count satisfies the linear recurrence given by the characteristic polynomial of the resulting 110×110 matrix. That bound is computed, not assumed.

5 The decision procedure

Let $c_1, \ldots, c_D$ be the coefficients of a claimed recurrence $a(n) = \sum_{i=1}^{D} c_i\, a(n-i)$, let $q(t) = t^D - \sum_{i=1}^{D} c_i t^{D-i}$ be its characteristic polynomial, and put

$$u_m \;=\; \tilde{w}^{\mathsf{T}} L^{m-1} q(L)\, \tilde{f} \qquad (m \geq 1).$$

Expanding the powers of L and using the displayed formula for a, one gets $u_m = a(n) - \sum_{i=1}^{D} c_i\, a(n-i)$ with $n = m + D$. So u_m is exactly the residual of the claim at that index, and the recurrence holds at an index n if and only if the corresponding u_m vanishes.

Now $u_m = \tilde{w}^{\mathsf{T}} L^{m-1} y$ with $y = q(L)\tilde{f}$ a fixed vector, so (u_m) satisfies the characteristic polynomial of L, of degree 110: writing that polynomial as $t^{110} - \sum_{i=1}^{110} \lambda_i t^{110-i}$ gives $u_{m+110} = \sum_{i=1}^{110} \lambda_i u_{m+110-i}$. Hence 110 consecutive vanishing residuals force every later residual to vanish, by induction. The test is therefore finite; and because it locates the *last* nonvanishing residual, it pins the threshold exactly rather than merely bounding it.

Everything is carried out in exact integer arithmetic on the vector $L^{m-1}y$. The matrix M is never formed.

6 Results

Theorem 1. *For A274957, the least order of a linear recurrence that a eventually satisfies is exactly* 81*, and that recurrence holds for every* $n > 83$*, the entry's own figure.*

Proof. The count is C-finite of order at most $S = 110$, so the minimal annihilator of the tail $(a(n))_{n \geq M}$ is the same for every M beyond S: the nilpotent part of L is killed after S steps. Taking such an M and 226 consecutive terms from there, the Berlekamp–Massey algorithm over $\mathbb{Q}$ returns the minimal linear recurrence generating them; since the sequence satisfies one of order at most S, $2S$ terms determine it. Its order is 81. That recurrence was then put through the residual test of Section 5: the last nonvanishing residual is at $n = 83$, followed by more than $S = 110$ consecutive vanishing ones, so it holds for every larger n. No recurrence of smaller order can hold eventually, since the tail's minimal annihilator is what the algorithm returned. □

7 Verification

Four checks were run, each reproducible from the code accompanying this note, and the first two are what pin the reading of the entry's English.

- **The published terms.** The walk count reproduces all 16 terms the entry publishes, exactly, in integer arithmetic, with the index taken from the entry's stated offset (1) rather than fitted. The first of them are 24, 648, 6936, 87780, 1119744, 14362920, 183551766, 2348053596, ...
- **An independent enumeration.** For $n = 1, \ldots, 1$ every one of the 3^{6n} arrays was written out and tested cell by cell against the condition as the entry states it — no transfer matrix, no row window, a separate piece of code. The counts agree with the walk count and with the entry.
- **The claim on the raw data.** Each claim was evaluated on the entry's published terms alone, with no model involved, wherever the proved range and the published data overlap. It holds there.
- **The annihilation test.** Carried to the derived bound $S = 110$ over the whole vector, in exact integer arithmetic, never sampled and never truncated.

8 References

1. OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A274957, revision 4, last modified Jul 12 2016.
2. R. P. Stanley, *Enumerative Combinatorics*, Vol. 1, 2nd ed., Cambridge University Press, 2012; §4.7, the transfer-matrix method.
3. M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011; C-finite sequences and their closure properties.

4. J. Berstel and C. Reutenauer, *Noncommutative Rational Series with Applications*, Cambridge University Press, 2011; linear representations of counting automata and their reduction.